

SSRD Reconstruction: Evaluation Summary Across All Tested Months

Seven ERA5 months (real ground truth) and four DART months (transfer test), Global / Europe / Germany / Korea

AWI Dark-Doldrums Project

2026-09-17

Scope

Eleven months have now been tested in total: seven on **ERA5** (real 3-hourly ground truth exists) and four on **DART** (only a real monthly mean exists; the 3-hourly reconstruction itself is what's being validated for). All eleven use the same production model (`models/model_kt.joblib`, trained once on ERA5 August 2025 / March 2026 / August 2026 – no winter data went into training) and the same evaluation code, broken down the same way: Global, Europe, Germany, Korea.

Two different metrics are used, and this report keeps them in fully separate sections rather than mixing them, because they answer different questions and behave very differently:

- **Pointwise R^2** – only possible on ERA5, since only ERA5 has real 3-hourly ground truth. Compares every individual 3-hourly prediction against the real value, at every grid cell. This is the time-series fidelity metric – does the model track *when* radiation rises and falls.
- **Monthly-mean spatial R^2** – the only metric possible on DART, and also computed on ERA5 for direct comparability. Aggregates to a monthly mean per cell first, then asks whether the model ranks *which cells* get more or less sun correctly. In a small region with little genuine spatial variance to explain, this statistic is much more fragile and can turn sharply negative even when the underlying predictions are good in an absolute sense.

Part 1: Pointwise R^2 (real 3-hourly ground truth, ERA5 only)

DART has no 3-hourly truth to compare against, so this section covers ERA5's seven tested months only.

region	Nov'25	Dec'25	Jan'26	Feb'26	Mar'26	Aug'25	Aug'26
Global	0.91	0.90	0.93	0.97	0.96	0.98	0.98
Europe	0.97	0.96	0.95	0.95	0.96	0.98	0.98
Germany	0.90	0.92	0.92	0.89	0.98	0.98	0.98
Korea	0.97	0.96	0.96	0.98	0.98	0.93	0.94

Reading: never below 0.89, for any region, in any of the seven months. This is the headline reassurance from the pointwise metric alone – the model’s grip on the actual time-varying signal, the part that determines whether a dark-doldrums event gets detected, is robust across the full annual cycle tested so far, in all three specific regions, including deep winter.

Part 2: Monthly-mean spatial R²

2a. ERA5 (all seven months)

region	Nov’25	Dec’25	Jan’26	Feb’26	Mar’26	Aug’25	Aug’26
Global	0.92	0.93	0.95	0.97	0.95	0.99	0.99
Europe	0.99	0.99	0.98	0.97	0.94	0.98	0.97
Germany	0.74	0.90	0.93	0.61	0.47	0.70	0.64
Korea	0.84	0.52	0.11	-1.31	-1.12	-0.07	-1.37

2b. DART (all four months – the only ground truth DART allows at all)

region	Jan’50	Apr’50	Jul’50	Oct’50
Global	0.98	0.95	0.98	0.97
Europe	0.97	0.87	0.97	0.98
Germany	0.87	0.41	0.53	0.92
Korea	0.43	-2.14	0.41	0.86

2c. Reading the spatial results together

Korea’s fragile spatial fit is a real property of the method, confirmed on real ERA5 ground truth, not manufactured by the ERA5-to-DART transfer. ERA5’s own Korea R² collapses from 0.84 (November) to -1.31 (February) – the same late-winter-into-spring window where DART’s Korea result is also weakest. This is the method’s own limitation, visible even with real ground truth and no transfer involved.

But the DART transfer clearly makes a bad month worse, not just “the same.” ERA5’s January Korea result is *marginal* (0.11) – not great, but nowhere near DART’s January (0.43) or, worse, DART’s April (-2.14, the single worst result found anywhere in this evaluation). DART’s own cloud scheme and snow/ice representation add a real, additional error on top of whatever spatial fragility the reconstruction method already has in that season.

A genuine gap this comparison exposes: DART’s worst result (April, Korea, -2.14) has no ERA5 cross-check yet – ERA5 has never been tested for April. Since April is DART’s worst month by a wide margin, ERA5 April is now the single most informative test left to run: it would show directly whether DART’s April crisis is “the known seasonal fragility, amplified,” matching the Nov-to-Feb arc already seen, or something specific to DART alone.

Germany shows the same qualitative shape, smaller amplitude. ERA5 Germany dips to 0.47-0.74 in its harder months (Mar, Aug, Nov) and recovers to 0.90-0.93 in Dec/Jan; DART

Germany dips to 0.41-0.53 (Apr, Jul) and recovers to 0.87-0.92 (Jan, Oct). The specific weak calendar months don't line up between the two datasets, which is itself informative: a shared month name (e.g. "January") doesn't guarantee matching synoptic weather between a reanalysis and a free-running climate simulation.

Practical implications

- **For time-series / event-detection work** (the actual dark-doldrums use case): Part 1's results support trusting the reconstruction across the full year, including winter, in all three specific regions.
 - **For anything relying on the absolute spatial pattern within a small region** in the transition months (roughly February through April, and to a lesser extent August): treat Korea's reconstruction with real caution; Germany's less severely but still noticeably.
 - **fa1 (surface albedo) remains the best-supported next step** – literature-validated as essential for this class of method, zero new downloads needed, and doubly motivated now: it explains both the DART transfer finding and this newly confirmed ERA5-native seasonal fragility.
 - **Test ERA5 April next**, specifically to resolve whether DART's worst result is "the known fragility, amplified" or a separate problem.
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All figures generated on Levante; source arrays and maps in `solar/data/reconstructed_ssrd/`, `solar/figures/`. Production model: `solar/models/model_kt.joblib`.